Investigation: COMPANHIA ENERGÉTICA DE BRASÍLIA - CEB (CVM code 14451), fiscal year 2010, as it was on 2013-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Energia Elétrica (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

Document used

2010 DFP, version 2, filed on 2011-07-18; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2011-07-15); 2 (2011-07-18).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=10177

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: component not computable: DEPI
- Alert by the study 1 rule: does not trigger alerts
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: —
- Percentile in group and year: —
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: low

DSRI, days' sales in receivables index
- Value: 1.076 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 77 (n = 51)
- Percentile in group and year: 68 (n = 461)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.016 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 40 (n = 51)
- Percentile in group and year: 66 (n = 471)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.966 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 30 (n = 52)
- Percentile in group and year: 42 (n = 475)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.125 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 74 (n = 51)

Percentile in group and year: 47 (n = 472)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: missing account: depreciation, cash flow lines in section 6.01.01
M-score component; it has no rule of its own
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low

SGAI, SG&A expenses index
Value: 1.026 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 54 (n = 51)
Percentile in group and year: 63 (n = 469)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.044 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 51 (n = 54)
Percentile in group and year: 42 (n = 498)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.005 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 54 (n = 53)
Percentile in group and year: 42 (n = 485)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 2.4 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 18.7 days)
Percentile in sector and year: 56 (n = 51)
Percentile in group and year: 60 (n = 443)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 9.10 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 79 (n = 51)
Percentile in group and year: 58 (n = 439)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.045 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 51 (n = 54)
Percentile in group and year: 43 (n = 491)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 0.167 (index)
Components: X1 = −0.084; X2 = −0.025; X3 = 0.039; X4 = 0.508 (indices)
Percentile in sector and year: 8 (n = 56)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 4/9 (5 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA (ΔROA): —
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): —
- Change in current ratio (ΔLIQUID): 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): —
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Highlighted in the following year's comparative: 2 (threshold of R$21.2 million, 1% of total assets of the version used). Values in R$ million:
DRE 3.01 Receita de Venda de Bens e/ou Serviços: 1,839.3 in the version used (v2, filed on 2011-07-18); 1,284.4 in the next DFP's comparative (v2, filed on 2012-04-09); difference of −554.9. Different value in the following year's comparative (restatement or reclassification).
DFC (direct method) 6.01 Caixa Líquido Atividades Operacionais: 113.2 in the version used (v2, filed on 2011-07-18); 86.4 in the next DFP's comparative (v2, filed on 2012-04-09); difference of −26.8. Different value in the following year's comparative (restatement or reclassification).
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller

differences) stay in the full list.
Full list: 54 accounts above the R$1,000 tolerance (group a, totals and key lines: 6; group b, reclassifications: 48), in the annex at the end.

What was known on the date

- FCA 2010, v1: 2010-03-22
- FCA 2010, v2: 2010-08-19
- DFP 2010, v1: 2011-07-15
- DFP 2010, v2: 2011-07-18
- DFP 2011, v1: 2012-03-31
- DFP 2011, v2: 2012-04-09

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Com Ressalva (qualified).
Excerpt checked literally: "Em nossa opinião, tais ajustes são apropriados e foram adequadamente efetuados."
Auditor in the 2010 FCA: KPMG Auditores Independentes.
Auditor in the 2011 FCA: KPMG Auditores Independentes.
Auditor in the 2012 FCA: KPMG Auditores Independentes.
Auditor changes: none in the FCAs filed by the date.
Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:30:33+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

Annex: full list of differences in the following year's comparative (R$ million)

(a) DRE 3.01 Receita de Venda de Bens e/ou Serviços: 1,839.3 in the version used, 1,284.4 in the comparative (difference of −554.9)
(a) DFC (direct method) 6.01 Caixa Líquido Atividades Operacionais: 113.2 in the version used, 86.4 in the comparative (difference of −26.8)
(a) BPP 2.03 Patrimônio Líquido Consolidado: 714.3 in the version used, 724.6 in the comparative (difference of 10.3)
(a) BPA 1 Ativo Total: 2,119.9 in the version used, 2,130.2 in the comparative (difference of 10.3)
(a) DRE 3.11 Lucro/Prejuízo Consolidado do Período: 19.6 in the version used, 18.8 in the comparative (difference of −0.8)
(a) BPP 2.01 Passivo Circulante: 627.9 in the version used, 627.9 in the comparative (difference of −0.0)
(b) DRE 3.04.02 Despesas Gerais e Administrativas: −1,057.6 in the version used, −129.3 in the comparative (difference of 928.3)
(b) DRE 3.04 Despesas/Receitas Operacionais: −1,112.3 in the version used, −211.2 in the comparative (difference of 901.1)
(b) DRE 3.03 Resultado Bruto: 1,196.0 in the version used, 295.1 in the comparative (difference of −900.9)
(b) DRE 3.02 Custo dos Bens e/ou Serviços Vendidos: −643.3 in the version used, −989.3 in the comparative (difference of −345.9)
(b) BPA 1.02.04.01.01 Contrato de Concessão: 386.1 in the version used, 244.7 in the comparative (difference of −141.4)
(b) DRE 3.04.05 Outras Despesas Operacionais: −0.0 in the version used, −82.8 in the comparative (difference of −82.8)
(b) DRE 3.04.04 Outras Receitas Operacionais: 0.3 in the version used, 55.0 in the comparative (difference of 54.7)
(b) BPA 1.02.02 Investimentos: 285.1 in the version used, 316.0 in the comparative (difference of 30.8)
(b) BPA 1.02.02.01 Participações Societárias: 285.1 in the version used, 316.0 in the comparative (difference of 30.8)
(b) DFC (direct method) 6.03 Caixa Líquido Atividades de Financiamento: −40.8 in the version used, −13.9 in the comparative (difference of 26.8)
(b) BPA 1.02.01 Ativo Realizável a Longo Prazo: 799.5 in the version used, 775.5 in the comparative (difference of −23.9)
(b) BPA 1.02.02.01.01 Participações em Coligadas: 283.4 in the version used, 302.6 in the comparative (difference of 19.2)
(b) DRE 3.02.01 Impostos e Contribuições: −643.3 in the version used, −661.1 in the comparative (difference of −17.7)
(b) BPA 1.02.01.03 Contas a Receber: 127.9 in the version used, 111.9 in the comparative (difference of −16.1)
(b) BPA 1.02.01.03.02 Outras Contas a Receber: 17.6 in the version used, 1.5 in the comparative (difference of −16.1)
(b) BPA 1.02.02.01.04 Outras Participações Societárias: 1.7 in the version used, 13.3 in the comparative (difference of 11.6)
(b) BPP 2 Passivo Total: 2,119.9 in the version used, 2,130.2 in the comparative (difference of 10.3)

(b) BPP 2.01.05.02.10 Outras Obrigações: 22.6 in the version used, 13.7 in the comparative (difference of −8.8)
(b) BPA 1.01.03 Contas a Receber: 321.2 in the version used, 329.1 in the comparative (difference of 7.9)
(b) BPA 1.01.03.01 Clientes: 321.2 in the version used, 329.1 in the comparative (difference of 7.9)
(b) BPA 1.01.03.01.09 Outros Créditos: 15.2 in the version used, 23.2 in the comparative (difference of 7.9)
(b) BPA 1.01 Ativo Circulante: 450.0 in the version used, 457.9 in the comparative (difference of 7.9)
(b) BPA 1.02.01.09 Outros Ativos Não Circulantes: 630.6 in the version used, 622.7 in the comparative (difference of −7.9)
(b) BPA 1.02.01.09.01 Ativos Não-Correntes a Venda: 286.9 in the version used, 279.0 in the comparative (difference of −7.9)
(b) BPP 2.03.09 Participação dos Acionistas Não Controladores: 224.3 in the version used, 231.6 in the comparative (difference of 7.3)
(b) BPA 1.02.03 Imobilizado: 197.4 in the version used, 192.9 in the comparative (difference of −4.5)
(b) BPA 1.02.03.01 Imobilizado em Operação: 197.4 in the version used, 192.9 in the comparative (difference of −4.5)
(b) BPP 2.01.05.02.01 Dividendos e JCP a Pagar: 5.3 in the version used, 9.7 in the comparative (difference of 4.4)
(b) BPP 2.03.05 Lucros/Prejuízos Acumulados: −82.5 in the version used, −79.5 in the comparative (difference of 3.0)
(b) BPA 1.02 Ativo Não Circulante: 1,670.0 in the version used, 1,672.4 in the comparative (difference of 2.4)
(b) BPP 2.01.05 Outras Obrigações: 134.0 in the version used, 131.7 in the comparative (difference of −2.3)
(b) BPP 2.01.05.02 Outros: 134.0 in the version used, 131.7 in the comparative (difference of −2.3)
(b) BPP 2.01.04 Empréstimos e Financiamentos: 154.2 in the version used, 156.5 in the comparative (difference of 2.3)
(b) BPP 2.01.04.01 Empréstimos e Financiamentos: 154.2 in the version used, 156.5 in the comparative (difference of 2.3)
(b) BPP 2.01.04.01.01 Em Moeda Nacional: 154.2 in the version used, 156.5 in the comparative (difference of 2.3)
(b) BPP 2.01.05.02.05 Encargos Regulatórios: 106.2 in the version used, 108.3 in the comparative (difference of 2.1)
(b) BPA 1.02.04.01.02 Diferido: 1.9 in the version used, 0.0 in the comparative (difference of −1.9)
(b) DRE 3.06.01 Receitas Financeiras: 46.6 in the version used, 44.8 in the comparative (difference of −1.8)
(b) DRE 3.06 Resultado Financeiro: −43.6 in the version used, −44.6 in the comparative (difference of −1.0)
(b) DRE 3.04.06 Resultado de Equivalência Patrimonial: 7.9 in the version used, 8.9 in the comparative (difference of 0.9)
(b) DRE 3.07 Resultado Antes dos Tributos sobre o Lucro: 40.1 in the version used, 39.3 in the comparative (difference of −0.8)
(b) DRE 3.09 Resultado Líquido das Operações Continuadas: 19.6 in the version used, 18.8 in the comparative (difference of −0.8)
(b) DRE 3.06.02 Despesas Financeiras: −90.2 in the version used, −89.5 in the comparative (difference of 0.8)
(b) DRE 3.11.01 Atribuído a Sócios da Empresa Controladora: 2.8 in the version used, 2.0 in the comparative (difference of −0.7)
(b) DRE 3.05 Resultado Antes do Resultado Financeiro e dos Tributos: 83.7 in the version used, 83.9 in the comparative (difference of 0.2)

(b) BPA 1.01.06 Tributos a Recuperar: 20.9 in the version used, 20.9 in the comparative (difference of −0.1)
(b) BPA 1.01.06.01 Tributos Correntes a Recuperar: 20.9 in the version used, 20.9 in the comparative (difference of −0.1)
(b) DRE 3.11.02 Atribuído a Sócios Não Controladores: 16.8 in the version used, 16.8 in the comparative (difference of −0.0)